

SCA AAD Cache Connector

The **SCA AAD Cache Connector** is an Azure Function App that bridges Microsoft Graph (Azure AD users) and the **Secure Contacts App (SCA)**. It automates user export, encrypted packaging, and blob storage delivery so SCA can consume prepared contact data without repeatedly querying Graph during each sync.

Who is it for?

Organizations that use Azure AD and want faster, controlled contact sync delivery for SCA while maintaining strong data-handling practices.

Key Value Propositions

- **Sync Efficiency:** Reduces repeated Graph paging during downstream sync operations by serving prepared exports from Blob Storage.
- **Security by Design:** Exports are written as encrypted password-protected ZIP files with tenant-specific key derivation context.
- **Flexible Authentication:** Supports both client secret mode and managed identity/default credential mode for Microsoft Graph access.
- **Operational Simplicity:** Supports scheduled and on-demand runs and can generate an SCA helper configuration JSON when enabled.

High-Level Architecture

1. **Trigger:** Azure Function runs on a timer schedule or HTTP request.
2. **Fetch:** Authenticates to Microsoft Graph using configured credential mode.
3. **Secure:** Transforms user data and writes encrypted ZIP exports.
4. **Store:** Uploads exports to Azure Blob Storage using atomic write behavior.
5. **Access:** SCA retrieves prepared export files from Blob Storage for sync consumption.

Core Requirements

- **Azure Subscription:** Required for Function App hosting and Blob Storage.
- **Azure Function App (.NET 8 Isolated):** Runs on Windows or Linux host.
- **Microsoft Graph Access:** Requires the documented Graph permission set and admin consent path.
- **Configuration and Secrets:** Environment settings are required; Key Vault references are recommended for sensitive values.

Get started in minutes

Provision the Function App and Storage Account, configure Graph access and environment settings, then trigger a run by timer or HTTP. The connector will produce encrypted exports that SCA can consume from Blob Storage.